

PROPOSED RESEARCH

Every second counts when a patient in need of an organ transplant is finally matched with a donor. The organ's post-transplant performance significantly declines as the time between organ removal and transplantation increases. Therefore, the transportation of organs from point A to point B must be executed as quickly and safely as possible to improve the chances of a successful transplant.

Drones have shown immense potential in delivering medical goods, vaccines, and life-saving organs to hard-to-reach locations. However, several challenges need to be addressed, particularly maintaining and monitoring container temperature and humidity. Drones carrying medical containers experience varying temperatures due to altitude changes, which may affect the organs being transported.

To address these challenges, this work presents the development of a smart, cost-effective medical container embedded with a Peltier module (thermoelectric cooler) and temperature sensor to maintain optimal storage conditions. A relay module is utilized to control the Peltier module, while the ESP8266 WiFi Microcontroller (MCU) enables real-time data transmission to the cloud, allowing remote monitoring and control of container temperature. The Blynk Internet of Things (IoT) platform is used for continuous temperature monitoring.

Results demonstrate that the proposed system accurately maintains the temperature according to user-defined values, ensuring the safety of sensitive healthcare products. For demonstration purposes, the container's internal temperature was maintained at 12°C, with performance tested in real-time at various altitude levels.

The primary goal of this project is to design a medical container using affordable materials and sensors, making it accessible for healthcare applications in India. While drones can be hired at a low cost (e.g., approximately ₹10,000), our focus is on creating an efficient, cost-effective container for transporting medical supplies such as blood and organs to areas where urgent medical care is required. The proposed solution offers a reliable, economical, and scalable approach to medical supply transportation, ultimately contributing to better emergency healthcare services across the country.

Our research integrates the concepts of medical drone delivery systems with the development of a smart container designed for Indian healthcare needs. The container is equipped with advanced sensors and materials to ensure temperature and humidity control, essential for organ and medical supply transportation. The cost-effective nature of the container design ensures affordability, making it feasible for widespread adoption in emergency healthcare services. By focusing on the container's design rather than the drone itself, the research prioritizes efficient medical supply delivery at minimal costs.

The container's design utilizes sensors that can be easily sourced and integrated, ensuring that the production cost remains low. The use of IoT technology through the Blynk platform enables real-time monitoring, providing healthcare professionals with the ability to track and manage the transportation process remotely. This system's scalability ensures that it can be implemented in various regions, especially in rural and hard-to-reach areas where medical supplies are often in high demand.

Additionally, the core focus of this research is the development of the software for the prototype design. The software system is responsible for managing sensor data, controlling the Peltier module, and transmitting real-time information to the cloud via the ESP8266 WiFi MCU. The software also ensures seamless interaction with the Blynk IoT platform, allowing users to monitor and adjust the container's temperature remotely. The design and implementation of this software aim to provide a user-friendly interface while maintaining high performance and reliability.

In conclusion, the proposed research aims to bridge the gap in medical transportation by introducing a smart, affordable, and efficient container system compatible with drone delivery services. By leveraging advanced sensor technology, IoT platforms, and custom-developed software, the system ensures safe and timely delivery of critical healthcare supplies, ultimately enhancing emergency medical services across India.

Transcript of 2nd Semester

As the 3rd semester result is still not published from university, I'm not able to submit it

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MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL
(Formerly known as West Bengal University of Technology)



PROVISIONAL GRADE CARD

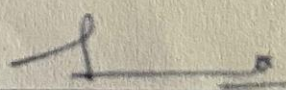
FIRST YEAR B.Tech. (IT) SECOND SEMESTER EXAMINATION OF 2023-24	
NAME : SHUBHADEEP DAS	ROLL NO. : 11700223094
REGISTRATION NO : 231170110537 OF 2023-24	
PROGRAM: BACHELOR OF TECHNOLOGY IN INFORMATION TECHNOLOGY	
COLLEGE / INSTITUTION: 117-RCC INSTITUTE OF INFORMATION TECHNOLOGY	

Subject Code	Subjects Offered	Letter Grade	Points	Credit	Credit Points
BSCH201	Chemistry-I (Gr-A)	D	5	4.0	20
BSM201	Mathematics - IIA	D	5	4.0	20
ESCS201	Programming for Problem Solving	D	5	3.0	15
HMHU201	English	B	7	2.0	14
BSCH291	Chemistry-I Laboratory (Gr-A)	O	10	1.5	15
ESCS291	Programming for Problem Solving	E	9	2.0	18
ESME291	Engineering Graphics & Design(Gr-A)	O	10	3.0	30
HMHU291	Language Laboratory	O	10	1.0	10
			Total	20.5	160

SGPA EVEN. (2nd) SEMESTER : - 6.67	
RESULT EVEN. (2nd) SEMESTER : P	

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Kolkata
16-07-2024


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